Literature Review

Project: Smart Clinic — an AI assistant that guides patients before they visit the doctor

Scope: Chest and respiratory conditions (symptoms, chest X-ray images, and X-ray reports)

Introduction

This review evaluates the existing work around Smart Clinic and uses it to refine the project. It covers three bodies of research that match the three components of the system: digital symptom checkers (Machine Learning), automated chest X-ray analysis (Deep Learning and Computer Vision), and radiology report processing (NLP). For each area we ask what is already known, what is missing, and what Smart Clinic will do differently. The review is organised around the five purposes of a literature review: context, research gap, justification, avoiding duplication, and conceptual framework.

1. Establishing Context & Foundational Knowledge

1.1 Symptom checkers and patient triage

Patients commonly search online before seeing a doctor. Semigran et al. (2015) audited 23 symptom-checker tools and found that the correct diagnosis was listed first in only about a third of cases, and appropriate triage advice was given in a little over half. Gilbert et al. (2020) later compared several symptom-assessment apps against general practitioners and found large differences in accuracy and safety between apps. The takeaway is that the need is real, but existing tools are inconsistent and mostly give a single generic answer from symptoms alone.

1.2 Deep learning for chest X-rays

Deep learning is now the standard approach for chest X-ray analysis. Kermany et al. (2018) trained CNNs on paediatric chest X-rays and released the dataset we use (normal vs. pneumonia). Rajpurkar et al. (2017) introduced CheXNet, a 121-layer DenseNet reported to detect pneumonia at a level comparable to radiologists on the ChestX-ray14 dataset (Wang et al., 2017). Irvin et al. (2019) released CheXpert with uncertainty labels. Transfer learning from networks such as ResNet (He et al., 2016) is widely used because medical datasets are small (Pan & Yang, 2010; Rahman et al., 2020).

1.3 NLP for radiology reports

Radiology reports are free text. Demner-Fushman et al. (2016) built the Indiana University (Open-i) collection that we use, with reports annotated by MeSH terms. Later work uses transformer models for report understanding, including ClinicalBERT (Alsentzer et al., 2019) and report labelers such as CheXbert (Smit et al., 2020). For Arabic users, AraBERT (Antoun et al., 2020) shows that Arabic text needs dedicated language models and preprocessing.

1.4 Explainability

Clinicians and patients are unlikely to trust a bare prediction. Grad-CAM (Selvaraju et al., 2017) highlights the image regions that influenced a CNN decision and is the common choice for medical images.

2. Identifying the Research Gap

Mapping the work above shows that each area is well studied on its own, but the patient's actual journey is not covered end to end.

Area	What is already known	What is missing
Symptom checkers	Symptoms can be matched to likely diseases; accuracy varies between tools.	Most tools work on symptoms only, are not tied to a specific specialty, and rarely support Arabic phrasing.
X-ray analysis	CNNs and transfer learning detect pneumonia with high accuracy; research is aimed at radiologists.	Results are rarely explained to patients in plain language, and models are presented as a standalone score.
Report processing	Reports can be classified and labelled automatically for research and clinical use.	Existing work serves clinicians. Little work explains report terms to the patient or links the report with symptoms and the image.
Integration	Each modality has strong models.	No simple, patient-facing flow combines symptoms, image, and report into one summary with a suggested specialty.

The gap Smart Clinic addresses: a single, patient-friendly tool that combines three sources of evidence (symptoms, X-ray image, X-ray report), explains them in simple language including Arabic symptom phrases, and ends with a suggested specialty and a downloadable summary to hand to the doctor.

3. Justifying the Project & Methodology

The literature supports the method choices in our proposal and also shows where they need to be improved. The table below lists the original plan, the improvement, and the reason from the literature. This is where the idea moves from the proposal to a realistic implementation.

Component	Original plan	Updated plan	Why
Symptoms (ML)	Compare Logistic Regression, Tree, Naive Bayes, KNN, SVM, Random Forest, Boosting.	Keep the comparison. Add a mapping from each disease to a specialty, focus evaluation on the respiratory subset, and show the top 3 results with probabilities.	The dataset covers 41 general diseases. Our scope is chest conditions, and the output must end with a specialty, not only a disease.
X-ray (DL/CV)	Simple CNN and Transfer Learning (ResNet/VGG).	Keep both. Add data augmentation, class weighting, a stratified re-split, and Grad-CAM heatmaps. Report confidence carefully.	Pneumonia images outnumber normal ones, and the dataset's original validation split is very small. Grad-CAM supports trust (Selvaraju et al., 2017).

Component	Original plan	Updated plan	Why
Reports (NLP)	TF-IDF + Logistic Regression and a Hugging Face model; term explanation.	Keep both. Create the normal/abnormal label from the report's MeSH annotations. Compare with a clinical or small transformer model. Add a medical glossary for term explanation.	IU X-Ray includes MeSH annotations that give reliable labels. Clinical language models perform better on medical text (Alsentzer et al., 2019).
Arabic layer	Dictionary + Regex.	Keep the dictionary and Regex. Add Egyptian-dialect phrases and basic text normalisation. Evaluate AraBERT only if time allows.	Arabic text needs dedicated handling (Antoun et al., 2020); the dictionary is fast and fits the timeline.
Integration	Show three results on one page.	Add a consistency check between modules (for example, symptoms and X-ray agree or disagree) and an urgent-symptom warning.	Combining evidence is the project's main contribution; safe handling of urgent cases reduces risk.
Tracking & cloud	MLflow, Azure ML, Streamlit.	No change.	These support reproducibility and deployment as planned.

What is new in Smart Clinic

- **Multi-source summary:** one workflow that joins symptoms, image, and report.
- **Specialty guidance:** disease possibilities are converted into a recommended specialty.
- **Patient-level explainability:** heatmaps on the X-ray and plain-language explanations of report terms.
- **Arabic support:** Arabic and Egyptian-dialect symptom phrases are understood.
- **Doctor-ready output:** a downloadable summary page the patient can show to the doctor.

Known limitations we will state openly

- The pneumonia dataset is paediatric and has only two classes, so the model cannot detect other chest conditions and may not generalise to adults. Results will be worded as "leans towards", not as a diagnosis.
- The symptom dataset is clean and structured. Very high accuracy on it may not reflect real patients, so we will analyse errors and avoid overstating performance.
- The three datasets come from different patients, so the modules are independent and are not trained on a shared case.
- Medical disclaimer: all outputs are preliminary guidance, not a final diagnosis and not a substitute for a doctor.

4. Avoiding Duplication of Effort

The literature shows what is already solved, so we do not repeat it:

- **X-ray models:** We will not design a new architecture or try to beat CheXNet-level results. We use established backbones (ResNet/VGG) and focus on a fair comparison with a CNN from scratch.

- **Data:** We will use the existing public datasets (Kaggle, Open-i) instead of collecting patient data, which avoids privacy issues and repeated data collection.
- **NLP models:** We use pre-trained Hugging Face models and compare them with a simple baseline instead of training a language model from scratch.
- **Where we spend time:** Our effort goes to what is not already available: the integration, the specialty mapping, the Arabic layer, and the patient-friendly explanation.

5. Conceptual Framework

From the literature we take the definitions, evaluation methods, and architecture for the project.

Part	Variables and methods taken from the literature	Evaluation
Mind 1: Symptoms	Binary symptom features (132) as input; disease label (41) as output; baseline then advanced models; PCA and feature selection.	Accuracy, Precision, Recall, F1; confusion analysis.
Mind 2: X-ray	Normalised and augmented images as input; normal or pneumonia as output; CNN vs. transfer learning; Dropout and Early Stopping.	Accuracy, Precision, Recall, F1, AUC; error analysis; Grad-CAM.
Mind 3: Reports	Cleaned text turned into TF-IDF or transformer embeddings; normal or abnormal label from MeSH; term extraction.	Accuracy, Precision, Recall, F1; compare baseline vs. transformer.
Integration	Outputs of the three models combined with the specialty mapping and urgent-symptom rules.	End-to-end testing with sample cases; MLflow experiment log.

Architecture: Patient input (symptoms, image, report) → three independent models → integration layer (specialty mapping, consistency check, warnings) → one summary page with the medical disclaimer.

Summary

The literature confirms that the problem is real and that the technical building blocks are mature. It also shows that no tool combines symptoms, X-ray image, and X-ray report for the patient in a simple and Arabic-friendly way. Smart Clinic does not try to invent new models; it contributes by integrating proven methods, improving them where the literature shows weaknesses (class imbalance, explainability, label quality), and presenting the result in a form the patient and doctor can use. The review leads to the tool updates in Section 3, which will be reflected in the next project document.

References

[1] Alsentzer, E., et al. (2019). Publicly available clinical BERT embeddings. Proceedings of the 2nd Clinical NLP Workshop.

[2] Antoun, W., Baly, F., & Hajj, H. (2020). AraBERT: Transformer-based model for Arabic language understanding. OSACT Workshop.

[3] Demner-Fushman, D., et al. (2016). Preparing a collection of radiology examinations for distribution and retrieval. Journal of the American Medical Informatics Association, 23(2), 304–310.

[4] Gilbert, S., et al. (2020). How accurate are digital symptom assessment apps for suggesting conditions and urgency advice? BMJ Open, 10(12), e040269.

[5] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. CVPR.

[6] Irvin, J., et al. (2019). CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison. AAAI.

[7] Kermany, D. S., et al. (2018). Identifying medical diagnoses and treatable diseases by image-based deep learning. Cell, 172(5), 1122–1131.

[8] Pan, S. J., & Yang, Q. (2010). A survey on transfer learning. IEEE Transactions on Knowledge and Data Engineering, 22(10), 1345–1359.

[9] Rahman, T., et al. (2020). Transfer learning with deep convolutional neural network for pneumonia detection using chest X-ray. Applied Sciences, 10(9), 3233.

[10] Rajpurkar, P., et al. (2017). CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning. arXiv:1711.05225.

[11] Selvaraju, R. R., et al. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. ICCV.

[12] Semigran, H. L., et al. (2015). Evaluation of symptom checkers for self diagnosis and triage: audit study. BMJ, 351, h3480.

[13] Smit, A., et al. (2020). CheXbert: Combining automatic labelers and expert annotations for accurate radiology report labeling using BERT. EMNLP.

[14] Wang, X., et al. (2017). ChestX-ray8: Hospital-scale chest X-ray database and benchmarks. CVPR.